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Sl. No. of Q.P.: 1568

Roll No.

Unique Paper Code : 2351403

Name of the Paper : Differential Equations - II (PDE & System of ODE)

F-8

Name of the Course : B.Sc. (II) Mathematics

Semester : IV

Duration : 3 Hours

Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

All the six questions are compulsory.

Attempt any two parts from each question.

1. (a) Find the partial differential equation arising out of the family of right circular cones whose axes coincide with the z-axis: $x^2 + y^2 = (z - c)^2 \tan^2 \alpha$.
(b) Define linear first order, quasi-linear partial differential equations with example. Give the geometrical interpretation of the same.
(c) Solve the initial-value problem: $3u_x + 2u_y = 0$, with $u(x, 0) = \sin x$. (12)
2. (a) Apply the method of separation of variables $u(x, y) = f(x)g(y)$ to solve the equation $u_x + u = u_y$, $u(x, 0) = 4e^{-3x}$.
(b) Derive two - dimensional wave equation.
(c) Transform the equation $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = 0$ into the canonical form and then find the general solution. (12)
3. (a) Obtain the solution of the Cauchy problem for the non-homogeneous wave equation:
$$u_{tt} = c^2 u_{xx} + h^*(x, t)$$

with the initial conditions:

$$\begin{aligned}u(x, 0) &= f(x), \\u_t(x, 0) &= g'(x).\end{aligned}$$

(b) Solve the initial-value problem:

$$\begin{aligned}u_{xx} + 2u_{xy} - 3u_{yy} &= 0, \\u(x, 0) &= \sin x, \\u_y(x, 0) &= x.\end{aligned}$$

(c) Determine the solution of the initial boundary-value problem:

$$\begin{aligned}u_{tt} &= 4u_{xx}, 0 < x < \infty, t > 0, \\u(x, 0) &= x^4, 0 \leq x < \infty, \\u_t(x, 0) &= 0, 0 \leq x < \infty \\u(0, t) &= 0, t \geq 0.\end{aligned}\tag{12}$$

4(a) Solve the initial boundary-value problem by the method of separation of variables:

$$\begin{aligned}u_{tt} &= c^2 u_{xx}, 0 < x < 1, t > 0, \\u(x, 0) &= x(1-x), u_t(x, 0) = 0, 0 \leq x \leq 1, \\u(0, t) &= u(1, t) = 0, t > 0.\end{aligned}$$

(b) Obtain the solution of the initial boundary-value problem:

$$\begin{aligned}u_t &= k u_{xx}, 0 < x < \pi, t > 0, \\u(x, 0) &= \sin^2 x, 0 \leq x \leq \pi, \\u(0, t) &= u(\pi, t) = 0, 0 \leq t < \infty.\end{aligned}$$

(c) Solve the Cauchy problem for the system:

$$\begin{aligned}u_{tt} - 9u_{xx} &= 0, x \in \mathbb{R}, t > 0, \\u(x, 0) &= \cos x, u_t(x, 0) = \sin 2x \text{ for } x \in \mathbb{R}.\end{aligned}\tag{13}$$

Q 5.(a) Use the operator method to find the general solution of the linear system of equations:

$$\frac{d^2 x}{dt^2} + \frac{dy}{dt} = t + 1, \quad \frac{dx}{dt} + \frac{dy}{dt} - 3x + y = 2t - 1.$$

(b) Find the particular solution of the linear system that satisfies the given initial conditions:

$$\frac{dx}{dt} = 6x - 4y, x(0) = 2, \quad \frac{dy}{dt} = x + 2y, y(0) = 3.$$

(c) Prove that the ordered pair of functions defined for all t by $(2e^{2t}, -3e^{2t})$ is a solution of

the Homogeneous linear system of equations:

$$\frac{dx}{dt} = 5x + 2y, \quad \frac{dy}{dt} = 3x + 4y.\tag{13}$$

Q6. (a) Using the method of successive approximations, find the sequence of functions that approaches the exact solution of the initial-value problem:

$$\frac{dy}{dx} = x^2 + y^2, y(0) = 1.$$

(b) Apply the modified Euler method to the initial-value problem:

$$\frac{dy}{dx} = x + y, y(0) = 0.$$

Apply the method to approximate the values of the solution y at $x=0$, and $x=1.0$ using $h = 0.5$.

(c) Apply the Runge-Kutta method to the initial-value problem:

$$\frac{dy}{dx} = 2x - y, y(0) = 3.$$

Apply the method to approximate the values of the solution y at $x=0.4$, using $h = 0.2$.

Carry the intermediate calculations in each step to five figures after the decimal points. (13)